Systems of Linear Equations : Direct Methods

Chapter 3 of Numerical Analysis

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Gaussian Elimination

Naive Gaussian Elimination

See the 3 ppts by John Carroll, Dublin City University

LU Decomposition for solving $A\mathbf{x} = \mathbf{b}$

In naive Gaussian elimination for solving $A\mathbf{x} = \mathbf{b}$, we apply row operations simultaneously to A and $\mathbf{b}$, that is, we left-multiply A and $\mathbf{b}$ with a suitable matrix M so that $U = MA$ is upper-triangular, and then solve $Ux = Mb$ by back-substitution. We do not store M, and we can only solve one equation by one set of matrix operations. There is another approach.

STEP 1: Decompose A as $A = LU$ where L is unit-diagonal lower-triangular and U is upper-triangular.

STEP 2: Solve $L\mathbf{y} = \mathbf{b}$, i.e. compute $\mathbf{y} = L^{-1}\mathbf{b}$.

STEP 3: Solve $U\mathbf{x} = \mathbf{y}$, i.e. compute $\mathbf{x} = U^{-1}\mathbf{y}$.

By doing so, we can solve an unlimited number of equations $Ax = \mathbf{b}$ with same A and different $\mathbf{b}$, by one set of matrix operations, which is LU decomposition.

LU Decomposition

$$\underbrace{\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}}_{A} = \underbrace{\begin{bmatrix} 1 & & & \\ l_{21} & 1 & & \\ \vdots & \vdots & \ddots & \\ l_{n1} & l_{n2} & \cdots & 1 \end{bmatrix}}_{L} \underbrace{\begin{bmatrix} u_{11} & u_{12} & \cdots & u_{1n} \\ & u_{22} & \cdots & u_{2n} \\ & & \ddots & \vdots \\ & & & u_{nn} \end{bmatrix}}_{U}$$

For $1 \le i, j \le n$,

$$a_{ij} = \sum_{k=1}^{n} l_{ik} u_{kj} = \sum_{k=1}^{\min(i,j)} l_{ik} u_{kj}.$$

Hence for $1 \le i \le j \le n$,

$$a_{ij} = \sum_{k=1}^{i-1} l_{ik} u_{kj} + u_{ij};$$

for $1 \le i < j \le n$,

$$a_{ji} = \sum_{k=1}^{i-1} l_{jk} u_{ki} + l_{ji} u_{ii}.$$

LU Decomposition Method 1: Doolittle Algorithm

Previously we have derived:

For $1 \le i \le j \le n$, $\quad a_{ij} = \sum_{k=1}^{i-1} l_{ik}u_{kj} + u_{ij}$.

For $1 \le i < j \le n$, $\quad a_{ji} = \sum_{k=1}^{i-1} l_{jk}u_{ki} + l_{ji}u_{ii}$.

This suggests the following algorithm:

Algorithm 3.1: Doolittle

Input: $A = [a_{ij}]_{n\times n}$

Output: $L = [l_{ij}]_{n\times n}$ and $U = [u_{ij}]_{n\times n}$ with L unit-diagonal and lower-triangular and U upper-triangular

```
for i in 1..n do
    for j in i..n do
        u_ij = a_ij − Σ_{k=1}^{i−1} l_ik u_kj
    for j in i + 1..n do
        l_ji = (a_ji − Σ_{k=1}^{i−1} l_jk u_ki) / u_ii
```

LU Decomposition Method 2: Standard Algorithm

Derive this algorithm in a recursive manner, but it nonetheless can be implemented with a loop. Partition A, L, U as

$$\underbrace{\begin{bmatrix} a_{11} & \boldsymbol{\mu} \\ \boldsymbol{\lambda} & A_{22} \end{bmatrix}}_{A} = \underbrace{\begin{bmatrix} 1 & \mathbf{0} \\ \boldsymbol{l} & L_{22} \end{bmatrix}}_{L} \underbrace{\begin{bmatrix} u_{11} & \boldsymbol{u} \\ \mathbf{0} & U_{22} \end{bmatrix}}_{U}.$$

Thus

$$\begin{bmatrix} a_{11} & \boldsymbol{\mu} \\ \boldsymbol{\lambda} & A_{22} \end{bmatrix} = \begin{bmatrix} u_{11} & \boldsymbol{u} \\ \boldsymbol{l} \cdot u_{11} & L_{22}U_{22} + \boldsymbol{l}\boldsymbol{u} \end{bmatrix}.$$

We then have

$$u_{11}, \boldsymbol{u} = a_{11}, \boldsymbol{\mu},$$
$$\boldsymbol{l} = \boldsymbol{\lambda} \cdot a_{11}^{-1},$$
$$L_{22}U_{22} = A_{22} - \boldsymbol{l}\boldsymbol{u}.$$

Properties of LU decomposition

1. Possibility of non-existence. It can be seen from the Doolittle algorithm that if any $u_{ii} = 0$, then the decomposition may not be able to proceed further. The exact condition for the existence can be found at `https://arxiv.org/pdf/math/0506382v1`
2. Non-uniqueness in pathological cases. If any $u_{ii} = 0$, then a matrix may have many LU decompisitions. For example,

$$\begin{bmatrix} 0 & 1 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ l & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & a \end{bmatrix}$$

   where only $l + a = 2$ is required.
3. Uniqueness in normal cases. If none of the u_{ii} is 0, then the LU decomposition is unique.

An observation on LU Decomposition and Gaussian Elimination

The multipliers used to zero out the entries below the pivot entry in Gaussian elimination are the negatives of the sub-diagonal entries in L. To see this, take for example $n = 3$. Let

$$A = \begin{bmatrix} 1 & & \\ l_{21} & 1 & \\ l_{31} & l_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ & u_{22} & u_{23} \\ & & u_{33} \end{bmatrix}.$$

Then

$$\begin{bmatrix} 1 & & \\ -l_{21} & 1 & \\ -l_{31} & 0 & 1 \end{bmatrix} A = \begin{bmatrix} 1 & & \\ 0 & 1 & \\ 0 & l_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ & u_{22} & u_{23} \\ & & u_{33} \end{bmatrix}.$$

$$\begin{bmatrix} 1 & & \\ 0 & 1 & \\ 0 & -l_{32} & 1 \end{bmatrix} \begin{bmatrix} 1 & & \\ 0 & 1 & \\ 0 & l_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ & u_{22} & u_{23} \\ & & u_{33} \end{bmatrix} = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ & u_{22} & u_{23} \\ & & u_{33} \end{bmatrix}.$$

That is,

$$\begin{bmatrix} 1 & & \\ 0 & 1 & \\ 0 & -l_{32} & 1 \end{bmatrix}\begin{bmatrix} 1 & & \\ -l_{21} & 1 & \\ -l_{31} & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & & \\ l_{21} & 1 & \\ l_{31} & l_{32} & 1 \end{bmatrix} = \boldsymbol{I}.$$

Equivalently,

$$\begin{bmatrix} 1 & & \\ 0 & 1 & \\ 0 & -l_{32} & 1 \end{bmatrix}\begin{bmatrix} 1 & & \\ -l_{21} & 1 & \\ -l_{31} & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & & \\ l_{21} & 1 & \\ l_{31} & l_{32} & 1 \end{bmatrix}^{-1}.$$

And this equation can be generalized to cases of arbitrary n.

Caution: Generally

$$\begin{bmatrix} 1 & & \\ -l_{21} & 1 & \\ -l_{31} & -l_{32} & 1 \end{bmatrix} \neq \begin{bmatrix} 1 & & \\ l_{21} & 1 & \\ l_{31} & l_{32} & 1 \end{bmatrix}^{-1}.$$

Example

Find the LU factorization of $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 2 \\ 3 & 1 & 5 \end{bmatrix}$**.**

Solution:

$$\begin{bmatrix} 1 & & \\ -2 & 1 & \\ -3 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 2 \\ 3 & 1 & 5 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & -4 \\ 0 & -5 & -4 \end{bmatrix}$$

$$\begin{bmatrix} 1 & & \\ 0 & 1 & \\ 0 & 5 & 1 \end{bmatrix}\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & -4 \\ 0 & -5 & -4 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & -4 \\ 0 & 0 & -24 \end{bmatrix}.$$

Hence

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 2 \\ 3 & 1 & 5 \end{bmatrix} = \begin{bmatrix} 1 & & \\ 2 & 1 & \\ 3 & -5 & 1 \end{bmatrix}\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & -4 \\ 0 & 0 & -24 \end{bmatrix}.$$

Gaussian Elimination with Partial Pivoting

Also see the 3 ppts by John Carroll, Dublin City University. But I will provide here a theoretical proof of the existence of a LU decomposition with row interchanges. To my best knowledge, a similar proof has not appeared in the literature.

Existence of LU decomposition with row interchanges

Theorem

For any square matrix A, there is a decomposition $PA = LU$ where P is a permutation matrix, L is a unit-diagonal lower-triangular matrix, and U is an upper-triangular matrix.

Proof.

We use mathematical induction on the order of the matrix. The theorem obviously holds when A is 1×1. For $n \geq 2$, suppose the theorem holds for any $(n-1) \times (n-1)$ matrix. Now, let A be $n \times n$. Consider two cases:

Case 1: The first column of A is all-zero. Then $A = \begin{bmatrix} 0 & \boldsymbol{\mu} \\ \mathbf{0} & \tilde{A} \end{bmatrix}$. Since $\tilde{A}$ is $(n-1) \times (n-1)$, the induction hypothesis says that there is $(n-1) \times (n-1)$ permutation matrix $\tilde{P}$, unit-diagonal lower-triangular matrix $\tilde{L}$, and upper-triangular matrix $\tilde{U}$, such that $\tilde{P}\tilde{A} = \tilde{L}\tilde{U}$. Let $P = \begin{bmatrix} 1 & \mathbf{0} \\ \mathbf{0} & \tilde{P} \end{bmatrix}$. Then

$$PA = \begin{bmatrix} 1 & \mathbf{0} \\ \mathbf{0} & \tilde{P} \end{bmatrix} \begin{bmatrix} 0 & \boldsymbol{\mu} \\ \mathbf{0} & \tilde{A} \end{bmatrix} = \begin{bmatrix} 0 & \boldsymbol{\mu} \\ \mathbf{0} & \tilde{P}\tilde{A} \end{bmatrix} = \begin{bmatrix} 0 & \boldsymbol{\mu} \\ \mathbf{0} & \tilde{L}\tilde{U} \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & \mathbf{0} \\ \mathbf{0} & \tilde{L} \end{bmatrix}}_{L} \underbrace{\begin{bmatrix} 0 & \boldsymbol{\mu} \\ \mathbf{0} & \tilde{U} \end{bmatrix}}_{U}.$$

Proof (continued).

Case 2: The first column of A is not all-zero. Let the $(d, 1)$ element of A be the chosen nonzero element, and let its value be a. Apply interchange between row 1 and row d. The corresponding permutation matrix is denoted by $\Pi_{1,d}$. Then $\Pi_{1,d}A = \begin{bmatrix} a & \boldsymbol{\mu} \\ \lambda & \tilde{A} \end{bmatrix}$. Since $\tilde{A} - \lambda\boldsymbol{\mu} \cdot a^{-1}$ is $(n-1) \times (n-1)$, the induction hypothesis says that there is $(n-1) \times (n-1)$ permutation matrix $\tilde{P}$, unit-diagonal lower-triangular matrix $\tilde{L}$, and upper-triangular matrix $\tilde{U}$, such that $\tilde{P}(\tilde{A} - \lambda\boldsymbol{\mu} \cdot a^{-1}) = \tilde{L}\tilde{U}$. Let $P = \begin{bmatrix} 1 & \mathbf{0} \\ \mathbf{0} & \tilde{P} \end{bmatrix} \Pi_{1,d}$. Then

$$PA = \begin{bmatrix} 1 & \mathbf{0} \\ \mathbf{0} & \tilde{P} \end{bmatrix} \Pi_{1,d}A = \begin{bmatrix} 1 & \mathbf{0} \\ \mathbf{0} & \tilde{P} \end{bmatrix} \begin{bmatrix} a & \boldsymbol{\mu} \\ \lambda & \tilde{A} \end{bmatrix} = \begin{bmatrix} 1 & \mathbf{0} \\ \tilde{P}\lambda \cdot a^{-1} & I \end{bmatrix} \begin{bmatrix} a & \boldsymbol{\mu} \\ \mathbf{0} & \tilde{P}(\tilde{A} - \lambda\boldsymbol{\mu} \cdot a^{-1}) \end{bmatrix}$$

$$= \begin{bmatrix} 1 & \mathbf{0} \\ \tilde{P}\lambda \cdot a^{-1} & I \end{bmatrix} \begin{bmatrix} a & \boldsymbol{\mu} \\ \mathbf{0} & \tilde{L}\tilde{U} \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & \mathbf{0} \\ \tilde{P}\lambda \cdot a^{-1} & \tilde{L} \end{bmatrix}}_{L} \underbrace{\begin{bmatrix} a & \boldsymbol{\mu} \\ \mathbf{0} & \tilde{U} \end{bmatrix}}_{U}.$$

So the theorem holds for any $n \times n$ matrix. □

Properties of LU decomposition with row interchanges

1. Assured existence.
2. Non-uniqueness. This is mainly because we often have many choices of selecting which row to change to the pivot position.

Gaussian Elimination with Scaled Partial Pivoting

See the 3 ppts by John Carroll, Dublin City University.

Scaled partial pivoting is equivalent to first scale the equations, i.e. make the scales of the equations equal or roughly equal to each other, and then do Gaussian elimination with partial pivoting.

Sensitivity Analysis of Linear Systems

Norms of vectors

How to qualify as a vector norm?

A vector norm is a real-valued function $\boldsymbol{x} \to \|\boldsymbol{x}\|$ defined over all real vectors that must have the following properties: For all real number α, vectors $\boldsymbol{x}$ and $\boldsymbol{y}$,

$$\begin{aligned} \|\boldsymbol{x}\| &\geq 0 \\ \|\boldsymbol{x}\| = 0 &\Leftrightarrow \boldsymbol{x} = \boldsymbol{0} \\ \|\alpha\boldsymbol{x}\| &= |\alpha|\|\boldsymbol{x}\| \\ \|\boldsymbol{x}+\boldsymbol{y}\| &\leq \|\boldsymbol{x}\| + \|\boldsymbol{y}\|. \end{aligned}$$

Norms of vectors (cont'd)

It is easy to see that the following are indeed qualified for vector norms:

Definition (common norms of a vector)

For a vector $\boldsymbol{x} = [x_1, x_2, \cdots, x_n]^T$, its **1-norm**, **2-norm**, and **∞-norm**, denoted by $\|\boldsymbol{x}\|_1$, $\|\boldsymbol{x}\|_2$, and $\|\boldsymbol{x}\|_\infty$, respectively, are defined, respectively, by

$$\begin{aligned}
\|\boldsymbol{x}\|_1 &= \sum_{i=1}^{n} |x_i| \\
\|\boldsymbol{x}\|_2 &= \sqrt{\sum_{i=1}^{n} x_i^2} \\
\|\boldsymbol{x}\|_\infty &= \max_{i=1}^{n} |x_i|.
\end{aligned}$$

Apparently, the above norms are all special cases of the generic p-norm: $\|\boldsymbol{x}\|_p = (\sum_{i=1}^{n} |x_i|^p)^{\frac{1}{p}}$.

Norms of matrices

How to qualify as a matrix norm?

A matrix norm is a real-valued function $A \to \|A\|$ defined over all real matrices that must have the following properties: For all real number α, matrices A and B,

$$\begin{aligned} \|A\| &\geq 0 \\ \|A\| = 0 &\Leftrightarrow A = \mathbf{0} \\ \|\alpha A\| &= |\alpha|\|A\| \\ \|A + B\| &\leq \|A\| + \|B\|. \\ \|AB\| &\leq \|A\|\|B\|. \end{aligned}$$

Norms of matrices (cont'd)

There are many choices for defining a matrix norm, but for numerical analysis, it is the most helpful to use one induced from a vector norm.

Definition (induced matrix norms)

For a matrix A, its **1-induced norm**, **2-induced norm**, and **∞-induced norm**, denoted by $\|A\|_1$, $\|A\|_2$, and $\|A\|_\infty$, respectively, are defined, respectively, by

$$\begin{aligned}
\|A\|_1 &= \sup_{\boldsymbol{x}\neq\boldsymbol{0}} \frac{\|Ax\|_1}{\|x\|_1} \\
\|A\|_2 &= \sup_{\boldsymbol{x}\neq\boldsymbol{0}} \frac{\|Ax\|_2}{\|x\|_2} \\
\|A\|_\infty &= \sup_{\boldsymbol{x}\neq\boldsymbol{0}} \frac{\|Ax\|_\infty}{\|x\|_\infty}.
\end{aligned}$$

It can be verified that they indeed qualify as matrix norms.

Norms of matrices (cont'd)

For computing matrix norms, we have the following theorem.

Theorem (computing matrix norms)

For an matrix $A = [a_{ij}]_{m\times n}$,

$$\begin{aligned}
\|A\|_1 &= \max_{1\le j\le n} \sum_{i=1}^{m} |a_{ij}| \\
\|A\|_2 &= \text{the largest singular value of } A \\
\|A\|_\infty &= \max_{1\le i\le m} \sum_{j=1}^{n} |a_{ij}|.
\end{aligned}$$

Sensitivity analysis of $A\boldsymbol{x} = \boldsymbol{b}$

Suppose that

$$A\boldsymbol{x} = \boldsymbol{b}.$$

Then $\boldsymbol{b}$ is perturbed a bit, to become $\boldsymbol{b} + \delta\boldsymbol{b}$, and let the perturbed solution be $\boldsymbol{x} + \delta\boldsymbol{x}$. That is,

$$A(\boldsymbol{x} + \delta\boldsymbol{x}) = \boldsymbol{b} + \delta\boldsymbol{b},$$

and therefore $A\delta\boldsymbol{x} = \delta\boldsymbol{b}$.

What is the relationship between $\|\delta\boldsymbol{b}\|$ and $\|\delta\boldsymbol{x}\|$? Let us first give a definition.

Condition number and its role in sensitivity analysis

Definition (condition number)

The **condition number** $\kappa(A)$ of a matrix A is defined to be $\|A\|\|A^{-1}\|$. It is dependent on the particular matrix norm used. If A is singular, then $\kappa(A) := \infty$.

Theorem (relative error bound by condition number)

Suppose that A is nonsingular and $\boldsymbol{b} \neq \mathbf{0}$. Then

$$\frac{\|\delta \boldsymbol{x}\|}{\|\boldsymbol{x}\|} \leq \kappa(A) \frac{\|\delta \boldsymbol{b}\|}{\|\boldsymbol{b}\|}$$

and there is some $\delta \boldsymbol{b}$ such that

$$\frac{\|\delta \boldsymbol{x}\|}{\|\boldsymbol{x}\|} = \kappa(A) \frac{\|\delta \boldsymbol{b}\|}{\|\boldsymbol{b}\|}.$$

Proof.

$\boldsymbol{b} \neq \boldsymbol{0}$ implies $\|\boldsymbol{b}\| \neq 0$, and combined with A being nonsingular, implies $\|\boldsymbol{x}\| \neq 0$. We have $\boldsymbol{b} = A\boldsymbol{x}$, thus $\|\boldsymbol{b}\| \leq \|A\|\|\boldsymbol{x}\|$, i.e. $\frac{1}{\|\boldsymbol{x}\|} \leq \frac{\|A\|}{\|\boldsymbol{b}\|}$. We also have $\delta\boldsymbol{x} = A^{-1}\delta\boldsymbol{b}$, thus $\|\delta\boldsymbol{x}\| \leq \|A^{-1}\|\|\delta\boldsymbol{b}\|$. Hence

$$\frac{\|\delta\boldsymbol{x}\|}{\|\boldsymbol{x}\|} \leq \|A\|\|A^{-1}\|\frac{\|\delta\boldsymbol{b}\|}{\|\boldsymbol{b}\|}.$$

On the other hand, there is some b such that $\|\boldsymbol{b}\| = \|A\|\|\boldsymbol{x}\|$, and there is some δb such that $\|\delta\boldsymbol{x}\| = \|A^{-1}\|\|\delta\boldsymbol{b}\|$. Therefore the equality can be attained. □

Remarks about condition number

- A condition number is always ≥ 1. It is 1 if A is some scalar multiple of the identity matrix.
- The specific value depends on the norm used, i.e., a matrix has different condition numbers with 1-norm, 2-norm, and ∞-norm. If the 2-norm is used, then $\kappa(A) = \frac{\sigma_{max}(A)}{\sigma_{min}(A)}$ where $\sigma_{max}(A)$ is the largest singular value of A and $\sigma_{min}(A)$ is the smallest singular value of A.
- It is an ideal measure of its singularity.
- A linear system $Ax = b$ is called **ill-conditioned** if it has a big condition number. It means small perturbation in $\boldsymbol{b}$ will result in big perturbation in the solution $\boldsymbol{x}$. This is beyond numerical methods.